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Fall 2026 **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

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Progress Summary

The pipeline runs end to end. Phase 1 (the learned and classical baselines) is done: Depth Anything V2 (DA-V2) is benchmarked on all 654 NYU-Depth v2 test images at three model scales (ViT-S/B/L) plus the metric indoor checkpoint, and the classical depth-from-focus (DFF) baseline (modified Laplacian focus measure, parabolic sub-frame interpolation, confidence map, all-focus guided smoothing) runs on the rendered focal stacks and all 120 DDFF 12-Scene test stacks (Hazirbas et al.). Phase 2 (the focus-informed metric refinement) is done on synthetic data: the robust scale fit and regularized local correction run on six scenes split into tuning and held-out sets, and the combined system runs qualitatively on real DDFF stacks. Preliminary ablations on the tuning scenes cover confidence weighting, the regularization and correction weights, and focal-stack size (2/3/5/10 frames).

Current Results

NYU-Depth v2 test split (654 images, Eigen crop), DA-V2 across model scales and alignment choices (Table 1):

Checkpoint	Alignment	AbsRel	RMSE (m)	δ_1
ViT-S	median scale only	0.494	2.349	0.393
ViT-S	least squares	0.062	0.297	0.956
ViT-S	robust IRLS	0.058	0.292	0.958
ViT-B	least squares	0.057	0.283	0.959
ViT-L	least squares	**0.055**	**0.277**	**0.961**
Metric-Indoor-S	none (raw metric)	0.218	0.650	0.670
Metric-Indoor-S	least squares	0.077	0.353	0.936

Table 1. DA-V2 by checkpoint and alignment on the NYU-Depth v2 Eigen test split; bold marks the column best.

The affine fit matters: a median scale alone collapses accuracy (δ_1 0.39), so we model $q = s\hat{d} + t$ instead of a scale-only alignment. Robust IRLS (iteratively reweighted least squares) buys a small, consistent gain over plain least squares. Bigger backbones help less than expected (S $\rightarrow$ L adds ≈ 0.005 δ_1 at $\approx 4\times$ the MPS runtime), so ViT-S stays the project baseline. The metric indoor checkpoint has a real scale prior but is biased on NYU (0.218 unaligned), and even aligned it trails the relative models.

Synthetic focal stacks (10 frames, focus 0.9–7 m, thin-lens blur model). Regularization weights were tuned on two scenes (gallery, stairs); Table 2 reports the held-out mean over the other four (corridor, objects, slanted, shelves), metric depth scored without alignment:

Method	AbsRel	RMSE (m)	δ_1
DFF raw	0.233	1.552	0.732
DFF + all-focus smoothing	0.221	1.510	0.702
DA-V2 aligned (oracle)	0.271	1.524	0.548
DA-V2 + scale fit from focus	0.308	1.614	0.459
Refined (ours)	0.259	1.483	0.575
Combined (ours)	**0.206**	**1.284**	0.704

Table 2. Held-out mean over the four eval scenes; combined falls back to the focus estimate on the rejected calibration (slanted).

The focus baseline is accurate where texture exists (δ_1 0.89–1.00 on textured regions) and degrades on flat surfaces, exactly where the confidence map reads low ($\bar{w} \approx 0.06$–0.26 textureless, 0.26–0.66 textured). All-focus guided smoothing recovers part of that loss (textureless AbsRel 0.77 $\rightarrow$ 0.61 on held-out scenes) but can smear confident edges into large flat regions, a known limitation of normalized convolution. On the calibrated held-out scenes (corridor, objects, shelves) the refinement has the lowest AbsRel of any non-oracle method on every split: all pixels 0.246 vs. raw 0.277, focus 0.266, scale fit 0.312; textured 0.211 vs. 0.224; textureless 0.281 vs. 0.307. Raw DFF keeps a higher δ_1 where texture exists (0.79 vs. 0.70) because its per-pixel maxima stay sharp there. The one AbsRel it loses is low-texture against oracle-aligned DA-V2 (0.281 vs. 0.265), and that fit uses ground truth. The *combined* system falls back to the focus estimate on rejected calibrations; per scene it matches the better component, and on the held-out mean it beats both (0.206 vs. 0.259 refined, 0.221 focus). DA-V2 underperforms the focus baseline on our scenes even with oracle alignment, plausible given how far the renderer sits from its training distribution, so we read this benchmark as a stress test rather than a real-data statement. Ablations on the tuning scenes: 2–3-frame stacks collapse the focus estimate (AbsRel 10.5–114.9 vs. 0.12–0.24 at 5–10 frames), and confidence weighting cuts both ways (refined AbsRel 0.188 vs. 0.210 uniform on stairs, but 0.164 vs. 0.114 on gallery).

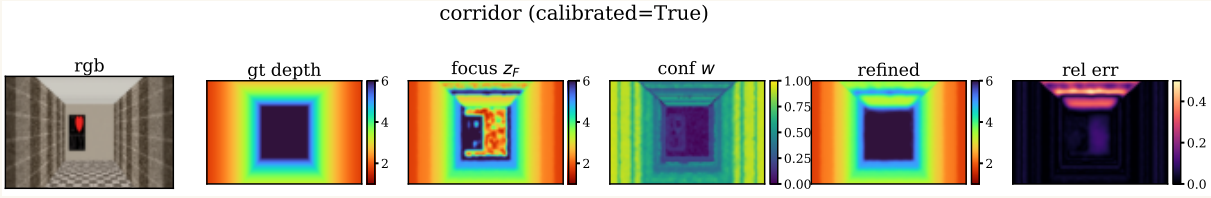


Fig. 1. Corridor, a held-out calibrated scene: input RGB, ground-truth depth, the focus estimate z_F, confidence w, our refined metric depth, and its relative error.

Difficulties

- **DDFF labels.** The public test archive ships 120 focal stacks but no ground truth (withheld for the benchmark server), and per-frame focus distances are not recorded. We reconstructed focus distances from the DDFF paper's refocus geometry (disparities linear in $[0.02, 0.28]$ px, $z \approx 0.14/d$), verified the frame-to-depth ordering per stack by Spearman correlation against the learned prior (about a third of the stacks we checked prefer a reversed ordering), and limited DDFF to qualitative evaluation. Confidences on the real Lytro stacks run low ($\bar{w} \approx 0.02$–0.05); a per-scene confidence normalization in the fit gets calibration accepted on most stacks, but none of it can be scored without labels.
- **NYU test split sourcing.** Both canonical sources were unreachable (the DenseDepth mirror

returns 403 and the NYU host refuses connections); we used a mirror of the same 654-image Eigen test split and verified image count, units, and depth range.

- **Defocus veiling.** In the layered renderer, blurred near-field texture bleeds over flat surfaces and lifts the focus measure there (the corridor's flat ceiling reads well above the textureless range). That is a genuine failure mode of the measurement, not a renderer artifact, and it caps how much we can trust w at occlusion boundaries.
- **Rejected fits.** Where the learned prediction anticorrelates with the focus estimate, the scale fit degenerates ($s \leq 0$). On those scenes the refined output falls back to a constant plane at the confidence-weighted focus mean, marked uncalibrated, rather than emit bad metric depth.

Changes from Proposal

DDFF 12-Scene moved from a quantitative to a qualitative dataset: the released test stacks carry no labels, as the proposal's planned verification found. Rendered scenes now carry the quantitative refinement evaluation, split into tuning and held-out sets so the reported numbers are not tuned on the held-out scenes. The proposal scheduled scale fitting and refinement for Oct 30–Nov 13; that stage already runs on the synthetic scenes, roughly three weeks ahead. Three additions beyond the proposed method: an all-focus guided smoothing stage on the DFF baseline (normalized convolution in inverse depth, weighted by confidence); a combined output that falls back to the focus estimate on rejected calibrations (best held-out mean, Table 2); and a degenerate-fit guard replacing silent bad calibration.

Next Steps

- Publish per-checkpoint runtimes on the webpage next to the NYU benchmark (the S/B/L and metric comparison is already up; runtimes are in the CSVs). Held-out evaluation and webpage work run Nov 14–24.
- Extend the DDFF qualitative evaluation to more stacks and, time permitting, phone-captured focal stacks as a second real source.
- Build the presentation deck (Dec 1–10) and keep the webpage figures in step with the final numbers.

References

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Code and figures: https://github.com/DimitriChrysafis/cs566-project